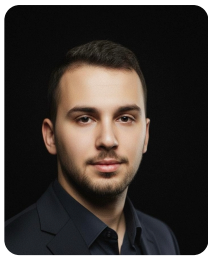
Muhammet Utku Küçük

Backend Software Engineer · Java & Spring Boot

Istanbul, Türkiye | +90 554 006 75 23 | utku@utkukucuk.com | utkukucuk.com |
linkedin.com/in/utkukucuk | github.com/utku-kucuk



SUMMARY

Backend Software Engineer with 3+ years of experience building, deploying and operating RESTful services and microservices with **Java and Spring Boot**, plus Node.js, Python and PHP. Delivered Web3 data integrations for a blockchain API company and backend services for 5+ corporate clients; containerized **10+ services** with Docker, cutting deployment time from ~2 hours to ~20 minutes. Skilled in API security (JWT, RBAC, LDAP), PostgreSQL and MongoDB data modeling, event-driven messaging with Kafka and RabbitMQ, Docker, Kubernetes and CI/CD with GitHub Actions.

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, SQL, PHP, Go, C#, C/C++

Backend: Spring Boot, Spring Security, Spring Data JPA, Hibernate, RESTful APIs, Microservices, Event-Driven Architecture, OpenAPI/Swagger, JWT, RBAC, LDAP/Active Directory, Node.js, Express.js

Databases & Messaging: PostgreSQL, MySQL, MSSQL, MongoDB, SQLite, Realm, Apache Kafka, RabbitMQ

DevOps & Cloud: Docker, Docker Compose, Kubernetes, AWS (EC2, S3), Linux, Nginx (reverse proxy, SSL/TLS), GitHub Actions (CI/CD), Cloudflare, Grafana

Testing & Tools: JUnit, Mockito, Testcontainers, Postman, Git, Maven, Agile/Scrum, Code Review

Frontend & IoT: React, Astro, Tailwind CSS; ESP32, ESPHome, MQTT, Home Assistant

EXPERIENCE

R&D Project Evaluator | Zaim Teknopark · Freelance · Remote — Oct 2026 – Present

- Perform technical due diligence on R&D project proposals from technology companies, assessing feasibility, innovation level and R&D eligibility.
- Review the architecture, technology stack and delivery plans of backend, cloud and IoT projects; audit progress and final reports and issue approval or revision recommendations.

Backend Developer | Covalent (CQT) · Contract · Remote — Sep 2025 – Feb 2026

- Built backend services on the Covalent Unified API, serving on-chain data (token balances, transactions, NFT metadata) from **10+ blockchain networks** to client applications.
- Delivered a token/NFT marketplace dApp end-to-end: REST API layer, wallet-based authentication, listing and transaction-history endpoints, and smart-contract interaction.
- Introduced response caching and request batching, cutting external API calls by **~40%** and average endpoint latency by **~35%**.
- Hardened integrations with pagination, input validation and rate-limit handling with exponential backoff; documented APIs with OpenAPI and shipped through PR-based code reviews.

Software Developer – Computer Vision | Nova UAV Team · University R&D Team · Istanbul — Nov 2024 – Apr 2025

- Developed a real-time image-processing pipeline in Python and OpenCV for UAV camera feeds: frame preprocessing, color segmentation and object detection.
- Optimized the frame-processing loop, raising throughput from **~15 to ~25 FPS** on onboard hardware.

Software Specialist – Backend & DevOps | Teknolojik Çözüm · Istanbul — Aug 2022 – Apr 2023

- Built backend services, REST APIs and MySQL-backed admin panels in PHP and JavaScript for **5+ corporate clients**.
- Containerized **10+ services** with Docker and Docker Compose, standardizing environments and reducing deployment time from **~2 hours to ~20 minutes**.
- Set up Nginx reverse proxies with SSL/TLS on Linux servers and CI/CD pipelines with GitHub Actions.
- Led log analysis and root-cause troubleshooting in production, reducing recurring incidents by **~30%** and sustaining **99.5%+ uptime**.

Software Development Intern | BİL Holding · Istanbul Oct 2020 – Jan 2021

- Built responsive UI components with HTML, CSS, JavaScript and Bootstrap and integrated them with backend REST APIs.

Software Development Intern | Kale Kalıp · Istanbul Mar 2020 – May 2020

- Implemented and unit-tested business-logic modules for internal applications; processed sensor data from production-floor IoT devices.

PROJECTS

Commerce Platform – E-Commerce Backend

Java 21, Spring Boot 3, Microservices, Spring Security, JWT, Spring Data JPA, PostgreSQL, Apache Kafka, Docker Compose, JUnit, Testcontainers

- E-commerce backend split into catalog, cart and order microservices with REST APIs documented in OpenAPI; stateless JWT authentication and role-based access with Spring Security.
- Event-driven order flow over Kafka, PostgreSQL per service, integration tests with Testcontainers, and the full stack running from a single Docker Compose file.

Viducate – Video Learning Platform

Live demo

Node.js, Express.js, PostgreSQL, Sequelize, FFmpeg, JWT, LDAP, Nginx, Docker, React

- Video courses, quizzes and progress tracking for three roles (Admin, Teacher, Student): large lecture uploads, HTTP byte-range streaming and FFmpeg thumbnail and duration extraction.
- JWT authentication with LDAP/Active Directory sign-in and just-in-time provisioning, CSV bulk user import and rate limiting; runs on Linux behind Nginx with Docker and PM2.

IronEcho – Offline-First Fitness Platform

Node.js, Express.js, MongoDB, Socket.io, Joi, JWT, React Native, Realm, Docker Compose

- Self-hosted REST API with JWT access/refresh tokens, Joi validation and real-time updates over Socket.io; a sync engine queues offline writes and resolves conflicts by timestamp, deployed with one Docker Compose command.

HireHub – Role-Based Job Board

Live demo

React 19, React Router 7, Context API, Tailwind CSS 4, Vite, GitHub Actions

- Job board for seekers, employers and admins with role-based route guards and an async, REST-shaped service layer over 1,000+ listings; linted, built and deployed by GitHub Actions.

utkukucuk.com – Personal Portfolio

Astro, TypeScript, Tailwind CSS, WebGL, GitHub Actions, Lighthouse CI, Cloudflare

- Bilingual (EN/TR) site with a custom WebGL hero, strict Content Security Policy and security headers; every pull request passes lint, type-check and Lighthouse CI before deployment.

EDUCATION

B.Sc. in Computer Engineering (English) | Istanbul Aydın University 2022 – 2025

Started at Bahçeşehir University, Computer Engineering (2020 – 2022); completed via lateral transfer.

Associate Degree in Computer Programming | Istanbul Aydın University 2018 – 2020

Information Technologies (Software & Networking) | Avcılar Anatolian Technical High School 2014 – 2018

Vocational training in programming and computer networks; IT internships at Kale Holding and Boğaziçi Döküman Çözümleri.

LANGUAGES

Turkish (Native) · English (B2 – Professional Working Proficiency)